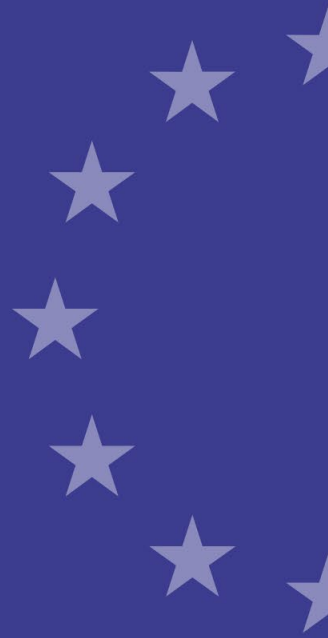




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Data Integration and VALIDation for SCiANTIX / SCiANTIX-DIVA

1. Abbreviations and Acronyms

Acronym	Full name	Category
CAGR	Commercial Advanced Gas-cooled Reactor	Nuclear reactor type
EOL	End-Of-Life	Physical quantity (time)
FGB	Fission Gas Behaviour	Physical phenomenon
FPC	Fuel Performance Code	Simulation tool
FRAPCON	Fuel Performance Code	Simulation tool
IFPE	International Fuel Performance Experiments	Literature database
MIT	Massachusetts Institute of Technology (License)	Licensing condition
MOX	Mixed Oxide Fuel	Material
NEA	Nuclear Energy Agency	Project
NM-KOS	Nuclear Materials Knowledge Organization System	Project / system
OECD	Organization for Economic Co-operation and Development	Project
PIE	Post-Irradiation Examination	Experimental technique
SCIANTIX	Fission gas behaviour module	Simulation tool
SEM	Scanning Electron Microscopy	Experimental technique
TEM	Transmission electron microscopy	Experimental technique
UO ₂	Uranium Dioxide	Nuclear fuel
V&V	Verification and Validation	Procedure

2. Overview and Rationale

Application Context and Domain

This case study is set within the framework of nuclear fuel performance modelling, with emphasis on the development, use and qualification of dedicated computer codes. The development of fuel performance tools is driven by the need to predict the complex behaviour of nuclear fuel during operation, which involves coupled thermal, mechanical, chemical, and neutronic phenomena (Van Uffelen et al., 2019). Within this context, fission gas behaviour (FGB) plays a relevant role to fuel performance, affecting fuel swelling and thermal conductivity, rod internal pressure and mechanical integrity (Rest et al., 2019).

To address FGB, SCiANTIX (Pizzocri et al., 2020; Zullo et al., 2023) has been developed as an open-source code designed to model FGB in nuclear fuel at the grain scale through computational modelling. SCiANTIX is designed to operate both as an independent tool and as an embedded module within industrial fuel performance codes (FPCs). Coupling SCiANTIX with FPCs improves flexibility thanks to its modular

structure and supports faster model development and validation within the tool itself. In line with established best practice, SCIANTIX and its coupling interfaces shall be developed, verified and validated (V&V) following rigorous quality procedures prior to their use in qualified contexts (Oberkampff & Roy, 2010).

Object and Scope of the study

The planned activity aims to structure SCIANTIX and its associated simulation data (inputs, outputs and validation-matrix databases) to meet the following objectives:

- (i) Data integration and code's flexibility: enhancing the code's ability to assimilate newly available data, e.g. those provided via OECD-NEA international benchmarks. This involves both input data (e.g., fuel properties (NEA, 2025)) and validation data (e.g., additional experiments available in literature or provided by dedicated projects as FUMEX (Boneva et al., 2013)), allowing for a systematic extension of the validation matrix;
- (ii) Standardized validation process: defining a formally rigorous separate-effect validation procedure for SCIANTIX. Separate-effect validation refers to the validation of individual physical mechanisms against selected experiments. The procedure will be formulated to support qualified use of SCIANTIX also by third parties, consistent with its open-source nature, and will include verification steps, definitions of figures of merit, and acceptability ranges.

To demonstrate the approach, the experimental dataset by White and coworkers (White, 2004; White et al., 2006) will be employed as a reference example for data integration and validation. The dataset provides a comprehensive characterization of porosity in irradiated UO_2 , linked to fission gas accumulation, and is therefore suitable for separate-effect validation of grain-scale FGB models.

Added Value and Innovation

The innovation relative to the state of the art is consistent with the two goals above:

- (i) While SCIANTIX modularity already allows some degree of assimilation of new experimental data by the development team, improved data structuring and integration workflows would facilitate SCIANTIX use by external users with fit-for-purpose datasets.
- (ii) Since validation of physical models is often performed on a case-by-case basis (e.g., (Cappellari et al., 2025)), this activity aims to define a standardized and reproducible separate-effect validation approach for an open-source code as SCIANTIX.

Material/s under investigation

The material under investigation is Uranium Dioxide (UO_2), a standard fuel for several reactor types. UO_2 represents the standard fuel matrix in SCIANTIX for modelling FGB. The UO_2 samples analysed in this case study were subjected to base-irradiation up to burnups around $21 \text{ GWd tUO}_2^{-1}$ in a Commercial Advanced Gas-cooled Reactor (CAGR), followed by power-cycle histories in the Halden Reactor. Extensive Post-Irradiation Examinations (PIE) for these samples were performed to study both intra-granular and inter-granular porosity, and are now available for separate-effect validation purposes. The fuel samples represent reference experimental material, sourced from the literature database compiled by White and coworkers (White, 2004; White et al., 2006) and made available for general use through the OECD Fuel Performance Database (*IFPE/CAGR-UOX-SWELL, Fuel Swelling Data Obtained from the AGR/Halden Ramp Test Programme*, n.d.).

Reference Research activity

Responsible organization: Politecnico di Milano (POLIMI).

Location: via G. La Masa, 34 – 20156 Milano, Building B12.

Involved persons			
Name	Surname	Contact email	Role
Davide	Pizzocri	davide.pizzocri@polimi.it	SST Leader, writing, conceptualization, supervision
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Giovanni	Zullo	giovanni.zullo@polimi.it	Contributor, software
Lelio	Luzzi	lelio.luzzi@polimi.it	Contributor, supervision

Reference Research project

The case study is developed only in the framework of NEO4MAT.

3. Computational Module

Simulation objective and scope

The simulation workflow involves the application of the SCIANTIX code to predict fission gas behaviour across multiple irradiation cases focused on grain-boundary bubble evolution, for which experimental data are available. *In this case study* the scope includes:

- (i) Data organization: input data and validation data for the various irradiation cases reported by White et al. (White, 2004; White et al., 2006) will be systematically organized. Data will be stored in dedicated folders, categorized by individual case study and structured to facilitate direct comparison with simulation outputs.
- (ii) Data generation: in addition to the existing input data, new datasets will be generated through the execution of SCIANTIX for each test case. This includes simulation data on gas concentrations, bubble evolution, and gas release but also supporting data such as verification reports, log files for reproducibility of the cases.

Computation objective and scope

The simulation activities focus on phenomenological modelling of FGB within nuclear fuel at the grain scale. The code addresses mesoscale phenomena occurring in nuclear materials during irradiation. It resolves physics-based mechanisms related to the time evolution of fission gases, from their production within the fuel grain, their diffusion within the grain, their accumulation in bubbles and subsequent release from the fuel matrix. Beyond modelling fission gas behaviour, the purpose of the simulations *in this case study* is to define and demonstrate a formally standardized and rigorous separate-effect validation process. The simulations are set up to reproduce the White and coworkers experimental case study, acting as a reference example for the definition of figures of merit and acceptability ranges for the validation of individual physical mechanisms. The case study serves also as a benchmark for the ongoing

restructuring of SCIANTIX. Specifically, it will verify the implementation of a new material library system, where properties are easily accessible to the user rather than being coded in the source files.

Software submodule

SCIANTIX is a mesoscale module employed to predict FGB in nuclear fuel at the grain scale via physics-based rate-theory models.

Computational Environment

SCIANTIX Version 2.2.1 is a multi-model simulation tool developed and maintained by Politecnico di Milano, Nuclear Engineering Division (POLIMI). The code is designed as a stand-alone module that can also be coupled with nuclear FPCs. SCIANTIX is open-source and released under the MIT licence.

In terms of accessibility, the code and all related data are open access and available through a public GitHub repository (*SCIANTIX Official GitHub Repository*, 2025).

In terms of execution, SCIANTIX follows an input-driven logic. It operates by reading external text files where the user defines the specific input parameters for the simulation. Also raw SCIANTIX outputs are handled via text files. Post-processed results, extracted from raw outputs and restructured for physical interpretation, are made available in tabular formats such as CSV, Excel or text files. Post-processing can be performed via Python scripts, which are also available in the GitHub repository.

In this case study simulations will be performed using SCIANTIX version 2.2.1, corresponding to the latest published version at the time of writing. This version follows SCIANTIX 2.0 (Zullo et al., 2023), and incorporates the advancements reported in recent publications. To ensure traceability and reproducibility throughout the project, the code version used in this activity will be frozen for the full project duration by creating a dedicated branch in the GitHub repository (*SCIANTIX Official GitHub Repository branch project/NEO4MAT_DIVA*, 2025). All project developments will therefore be based on this branch, avoiding interference with parallel ongoing developments in SCIANTIX. Simulations will be executed on a Linux-based virtual machine.

Software Validation context

SCIANTIX has been validated against separate-effect experiments and PIE data available in the open literature (Zullo et al., 2023). For separate-effect validation, the code relies on experimental data available in the literature covering: intra-granular and inter-granular swelling experiments, Helium behaviour under annealing, fission gas release measurements, high-burnup structure characterization.

Following its coupling with several FPCs, SCIANTIX has been validated also at the integral scale of the fuel rod (Zullo et al., 2024). For integral validation, the coupled code is compared against experimental datasets from the International Fuel Performance Experiments (IFPE) database, with particular reference to integral fission gas release.

Software Uncertainty and Limitations (Qualitative)

The primary uncertainties in SCIANTIX arise from the high sensitivity of its mechanistic models to material parameters such as diffusion coefficients, bubble nucleation and resolution rates, and surface tension values. These parameters are either derived from lower-length scale calculations or from indirect experimental measurements, both of which carry intrinsic uncertainties. *In addition, for the specific case study*, when

comparing simulation outputs to the experimental data, it must be acknowledged that the PIE measurements carry their own intrinsic experimental uncertainties and measurement errors.

Additionally, being an input-driven engine, the reliability of the outputs depends on the accuracy and resolution of the provided irradiation history and initial conditions. In the specific case study, the primary simulation uncertainty lies in

- (i) Translating the irradiation history (i.e., the base irradiation in the Advanced Gas-cooled Reactor and the subsequent power-cycles in the Halden Reactor) into 0D input parameters.
- (ii) Defining the exact initial state of the specific samples, e.g. distribution in initial grain size within the physical fuel pellet might not be perfectly captured by a single nominal value.

Regarding numerical uncertainties, these arise from the mathematical discretization and algorithmic solving of physical equations, such as the solution of the intra-granular diffusion of fission gases.

The code intrinsic limitations include:

- (i) SCIENTIX is a zero-dimensional (0D) tool; it assumes spatial homogeneity within the representative volume of the grain and cannot independently simulate macroscopic fuel rod behaviour (e.g., radial temperature gradients, pellet-cladding interaction) unless coupled online with an integral FPC.
- (ii) When coupled with an FPC, limitations may occur if the FPC does not directly provide the local inputs required by SCIENTIX. For example, local hydrostatic stress is not calculated in FRAPCON (Pizzocri et al., 2025).
- (iii) While extensively validated for xenon, krypton and helium in UO_2 , the code requires validation for other fuel types (e.g. MOX, MA-bearing fuels), and chemically active volatile fission products (e.g., iodine caesium, tellurium).
- (iv) As a mesoscale tool focused on the grain scale, a current structural limitation lies in directly connecting these models with lower-scale simulations (e.g., atomistic, dislocation dynamics), and with the description of the material's micro-structure evolution (e.g., fuel grain restructuring, grain shape change).

Input Parameters and Boundary Conditions

The simulation requires the following specific input data:

- Settings (input_settings.txt) for model selection and solver configuration: a single-column text file with integer flags selecting which physics models and numerical methods to use. Each line contains an integer flag and a comment explaining the available options.
- Boundary conditions, irradiation history (input_history.txt) for time-dependent conditions: a four-column text file defining the simulation history with linear interpolation between specified points. The code automatically interpolates between these rows using a fixed number of time steps. The four columns represent time in hours, local temperature in Kelvin, local fission rate density in fissions $\text{m}^{-3} \text{s}^{-1}$, and local hydrostatic stress in MPa.
- Initial conditions (input_initial_conditions.txt) for physical parameters and initial state: a structured text file with values followed by comments. Each data line is followed by a comment line (starting with #) describing the parameters. This file

provides the code with the initial state of the material, including fabrication characteristics such as the initial grain radius and initial porosity.

- Scaling factors (input_scaling_factors.txt) for parameter adjustment: a single-column text file with scaling factors applied to model parameters for sensitivity/uncertainty purposes. It is an optional file. Details on available options are avoided since out of the scope of the case study.
- Material library with matrix, gas and system (i.e., gas within the matrix) properties, currently stored in source files of the code (.C). The stored properties are those required for the physical models implemented in SCIANTIX.

For this case study, all relevant experimental input parameters and irradiation histories are sourced directly from the experimental database by White and coworkers.

Physical models descriptions

SCIANTIX utilizes physics-based models integrated within a multi-scale modelling approach. Below is a table listing the physical models implemented in SCIANTIX 2.2.1 relevant for the case study; its primary focus is the behaviour of noble gases and the evolution of the fuel microstructure at the grain scale.

Model	Category	Target/description	Main assumptions	Reference
Densification	Microstructure, semiempirical	Burnup driven fuel densification, represented as a progressive reduction of the fabrication porosity towards a residual porosity.	Empirical fit, temperature and burnup dependent only;	(Van Uffelen, 2002)
Grain growth	Microstructure, physics-based model	Grain radius growth during irradiation as a function of temperature, burnup and grain-boundary mobility.	Pinning effect of secondary phases is neglected.	(Ainscough et al., 1973)
Gas production	FGB, physics-based model	Production of gaseous species from fission, calculated via local fission rate and gas fission yields.	Constant fission yields;	Standard fission gas production law.
Gas diffusion	FGB, physics-based model	Intra-granular fission gas diffusion solving for gas concentration in intragranular solution and in intragranular bubbles.	Spherical grains with uniform temperature; quasi-steady-state between gas atoms and bubbles.	Standard diffusion law using a spectral (Booth-type) formulation.
Intragranular bubble behaviour	FGB, physics-based model	Intragranular bubble population, atoms per bubble, bubble radius, and intragranular gaseous swelling. Intragranular bubble	Stationary bubbles, uncertainties in bubble nucleation, trapping of	(Pizzocri et al., 2018; White & Tucker, 1983)

		concentration evolved with a simple source–decay law; bubble radius/volume inferred from the gas inventories stored in the intragranular bubbles.	gas atoms and resolution models.	
Intergranular bubble behavior	FGB, physics-based model	Intergranular bubble evolutions: bubble volume/radius/area, fractional coverage, bubble pressure, intergranular gas swelling, and released inventories.	Standard bubble shape,	(Pastore et al., 2013)
Grain-boundary micro-cracking	FGB, physics-based model	Degradation (micro-cracking) of grain-boundary integrity during transients and progressive healing with burnup.	Unique bubble size, pressure, and shape; grain-boundary healing is empirically derived from burnup.	(Barani et al., 2017; Cappellari et al., 2025)
Grain-boundary venting	FGB, physics-based model	Venting of gases from grain-boundary edges to the free volume, given the fuel open porosity.	Strongly dependent on densification model.	(Claisse & Van Uffelen, 2015)

The code includes standard modules for burnup, radioactive decay and microstructure updates. Not relevant for the case study but available in the codes:

- (i) Chromium solubility in UO_2 as a function of temperature and oxygen potential.
- (ii) Gap partial pressure in the fuel-cladding gap based on thermodynamic equilibrium relations. The intent is to represent oxidizing conditions in the gap atmosphere when stoichiometry deviation effects are activated.
- (iii) Stoichiometry deviation of nuclear fuel under oxidising conditions.
- (iv) Grain-boundary sweeping of intragranular gas towards the grain boundary caused by grain growth.
- (v) Intragranular bubble diffusivity, which is relevant at high temperatures.
- (vi) Effective burnup that accumulates only when the fuel temperature is below a threshold. The intent is to represent burnup-driven effects that are assumed to be active primarily at relatively low temperature.
- (vii) High-burnup structure formation in terms of a burnup dependent phase transformation law, porosity evolution and related pore statistics.

Additional details are reported in SCIANTIX documentation (SCIANTIX Official documentation, 2025), where references for each model are reported.

Simulation Workflow and Execution

The simulation workflow consists of the following steps:

1. Setup of the required input files for each case in the experimental database.
2. Execution of SCIANTIX in each working directory.

- a. Initialization: read input files, create and initialize simulation objects as fuel matrix, gas species, and systems, set up numerical solvers and time integrators.
 - b. Gas behaviour calculations: solves gas behaviour accordingly to the selected models.
 - c. Microstructural updates in fuel matrix properties, porosity and high-burnup structure evolution.
 - d. Environmental interactions as temperature, irradiation condition and mechanical changes.
 - e. System evolution in time, variables update.
 - f. Output generation at specified time steps with fuel properties and gas behaviour predictions.
3. Production of raw numerical outputs and comprehensive log files that detail the execution process and verify code performance.
 4. Post-processing of the raw outputs via Python scripts for direct comparison with experimental PIE results.

The simulation runs will execute the irradiation histories provided in the White database for the selected experimental samples, to allow for comparison with the experimental PIE data. Parameter sweeps are not the primary focus of this case study.

Simulation Data Output Planned and Their Role

The data generated by SCIANTIX are categorized into primary simulation data and supporting data. The primary outputs consist of the raw numerical data organized in specific output text files (output.txt) that record the temporal evolution of key variables, including intra-granular and inter-granular gas concentrations, bubble density, and localized fuel swelling. These outputs provide the predicted evolution of the fuel. They shall be compared to the experimental PIE measurements to perform a quantitative separate-effect validation of the code.

Beyond these physical results, the code architecture will include a quality assurance protocol that produces a report together with the primary simulation outputs.

First of all, a pre-execution verification phase is foreseen. This includes a suite of non-regression tests to ensure that any recent updates to the source code have not affected the performance of existing models. By comparing current results against a set of gold reference cases, the system confirms that the code version remains stable. Furthermore, SCIANTIX will incorporate automated verification checks that monitor whether the simulation stays within the established physical boundaries of the adopted models. If unphysical parameters or irradiation histories are detected, the system is designed to trigger error reports.

Numerical stability and accuracy are also explicitly addressed providing a feedback on potential errors due to mathematical discretization, such as the impact of the timestep length or the discretization in the diffusion solver.

Lastly, the system will automatically produces log files that echo the initial inputs and settings ensuring results reproducibility.

Data output post-processing (if applicable)

The primary purpose of the post-processing is to extract specific time-steps (typically the End-Of-Life, EOL) from the simulation output and manage them for direct comparison with the experimental PIE measurements.

Data extraction and processing are performed via dedicated Python scripts. The raw text outputs are parsed and converted into structured tabular formats (e.g., CSV, Excel). Parity plot are employed for direct comparison with experimental data (predicted vs. measured values). This graphical interpretation allows to directly

visualize the model's accuracy, quantifying the scatter and deviations of the identified figure of merits from the ideal 1:1 agreement line.

4. Literature dataset module

Literature-based datasets scope and objective

The scope of this literature dataset encompasses PIE measurements of inter-granular porosity development in irradiated UO_2 fuel from the work of White and coworkers (White, 2004; White et al., 2006). The datasets document the evolution of fission gas bubbles on grain faces, covering processes such as bubble growth, coalescence, morphological relaxation, and venting to grain edges under varying transient conditions.

Strategic relevance

This dataset is relevant due to the influence of grain-face bubble behaviour in understanding whether fission gases are retained within the fuel matrix or released. Accurate modelling of these phenomena is essential for fuel performance assessments, as retained gases can cause fuel swelling while released gases can deteriorate fuel thermal performance and increase pressure within the fuel rod. By utilizing this comprehensive experimental basis, the study aims to provide a robust validation matrix for the SCIANTIX code in the context of oxide fuels.

Selection criteria

This specific dataset was selected given its high level of detail and statistical significance. It includes swelling measurements extracted from multiple radial locations per fuel sample, Transmission Electron Microscopy (TEM) and Scanning Electron Microscopy (SEM) micrographs. The data includes exact numerical measurements for grain-face area, bubble density, fractional coverage by area, volumetric swelling, mean projected bubble radius, mean projected bubble length, and the fraction of vented pores. Furthermore, it covers a wide variety of operational scenarios, including fast ramps, slow ramps, and power cycling, with local fuel temperatures ranging from 1200 to 1920°C. The physical processes of swelling and gas release documented in this dataset are considered representative for fuel up to burnups of 45-50 GWd/tU (White, 2004).

Dataset identification

The dataset is extracted from the peer-reviewed scientific article "The development of grain-face porosity in irradiated oxide fuel" authored by R.J. White and published in the Journal of Nuclear Materials (2004). The specific experimental data originate from British Energy campaigns involving CAGR UO_2 fuel, which was base-irradiated to burnups of approximately 21 GWd/tU and subsequently ramped in the Halden Reactor. The data have also been made available for general use through the OECD Fuel Performance Database, where it is identified as NEA-1705 (IFPE/CAGR-UOX-SWELL) (IFPE/CAGR-UOX-SWELL, *Fuel Swelling Data Obtained from the AGR/Halden Ramp Test Programme*, n.d.) and compiled by British Energy and available through the OECD-NEA Data Bank.

Role and purpose

Within the context of the case study, the role of this dataset is to serve for the separate-effect validation matrix of SCIANTIX.

Planned usage

The dataset will be used in two distinct phases of the simulation workflow. First, the documented power histories will be extracted to provide the driving boundary conditions for the simulations. Second, the summary of derived swelling data will be used as the target metrics to directly compare against the simulation outputs (code-to-data comparison).

5. Integration of literature and simulation data

Rationale for data integration (contextualization and strategic objectives)

SCIANTIX simulations aim at predicting fuel behaviour along with irradiation via physics-based models. To validate model performances simulation outputs shall be compared with experiments. Therefore, experimental activities represented by the reference to the White database serve as a validation matrix for the code results, thanks to code-to-experiment comparison. They provide both the boundary conditions for the preparation of the input files and the final comparison results to validate the code and its models.

Integration workflow

The planned data flow operates in a two-step mechanism. First, there is a one-way flow from the experimental database to the simulation environment to setup simulation input files. Second, once the simulation is executed, a comparative data flow occurs: the continuous simulation outputs are post-processed to extract specific metrics exactly at the EOL time-steps. These simulated metrics are then aligned and compared directly against the experimental PIE values to formally assess the code's predictive accuracy.

Limitations (Qualitative)

Achieving consistency between experimental observations and deterministic simulations requires acknowledging several inherent limitations. The most significant assumption involves the spatial translation of the environment: the macroscopic, three-dimensional experimental fuel samples must be mapped into the zero-dimensional (0D) framework of SCIANTIX. This is addressed by treating each radial sampling location from the White dataset as an independent 0D representative volume element subjected to local boundary conditions.

Furthermore, there is an intrinsic gap between the deterministic nature of the simulation and the stochastic nature of PIE data. Experimental measurements are subject to local microstructural heterogeneities, measurement scatter, and uncertainties. These uncertainties must be considered when evaluating the code's predictions in the final parity plots, as perfect agreement is often precluded by the inherent variance in the physical samples.

6. Planned Data Outputs and File-Level Organisation

The simulation module is designed to produce a comprehensive set of results for each experimental sample identified in the White database. These outputs are categorized into raw numerical data, which capture the full temporal evolution of the fuel, and post-processed datasets specifically curated for the validation phase. The primary outputs are generated as standard plain text files (.txt), these raw files are parsed into structured tabular formats, such as CSV or Excel (.xlsx) or again text files, which

contain the extracted Figures of Merit at critical timestamps. Additionally, the post-processing pipeline generates graphical representations, such as parity plots, stored as images (.png) for direct inclusion in technical reports.

Files will be organized following a strict hierarchical principle. The directory structure will be centred on the specific experimental sample ID (e.g., /simulations/sample_XXXX/), ensuring that all input parameters, execution logs, raw outputs, and final post-processed results for a given test case are co-located. This hierarchical organization is essential for supporting the project's quality assurance goals.

Activity	Planned Dataset	
	Planned dataset category	Simulation input files
Defining input files for the simulation	Dataset description	Nuclear fuel properties, initial conditions, irradiation history, available physical models activated in the simulation.
	Expected file types	Tabular
	Planned file formats	Tabular/Text: Standard inputs (.txt). Header files: Material library properties (.h or .C).
	Organisation principle	By sample ID
	Indicative data volume	< 10 KB
Simulations execution	Planned dataset category	Simulation raw output files
	Dataset description	Evolution of all calculated physical variables. Includes automated reports on code verification, non-regression tests, numerical errors, and acceptability ranges.
	Expected file types	Tabular, text
	Planned file formats	Tabular/Text/Log: .txt for data; .log, .md, or .html for verification reports.
	Organisation principle	By sample ID
	Indicative data volume	< 300 KB
Post-processing	Planned dataset category	Post-processed output files
	Dataset description	Numerical data extracted from raw outputs and restructured for physical interpretation.
	Expected file types	Tabular, images
	Planned file formats	Tabular/Images: .csv, .xlsx, .json, or .txt. Graphical plots in .png
	Organisation principle	By sample ID
	Indicative data volume	< 300 KB
Separate-effect validation	Planned dataset category	Literature dataset
	Dataset description	Experimental data collected in open literature.
	Expected file types	Tabular, images
	Planned file formats	Tabular formats (CSV), Excel, JSON or text files, .png
	Organisation principle	By sample ID, or unique reference folder
	Indicative data volume	< 10 KB

7. Datasets visibility and licensing conditions

All datasets generated by this simulation module will be made publicly visible in the SCIENTIX GitHub repository. The simulation tool and associated data are already accessible under MIT open-source license via its official GitHub repository.

8. Meaning and Intended Use of the Data Output

The simulation outputs represent the physics-based time evolution of mesoscale fission gas behaviour (e.g., intra/intergranular concentrations, bubble density, volumetric swelling) calculated over a 0D equivalent fuel grain. The primary intended use of these datasets is to establish a quantitative and reproducible baseline for the separate-effect validation of SCIANTIX. By formalizing this procedure, the study contributes to a standardized validation framework for the code. Furthermore, the structured outputs are specifically formatted to ensure full integration into the NM-KOS (Nuclear Materials Knowledge Organization System) framework.

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